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10.2.1 A simple QR algorithm

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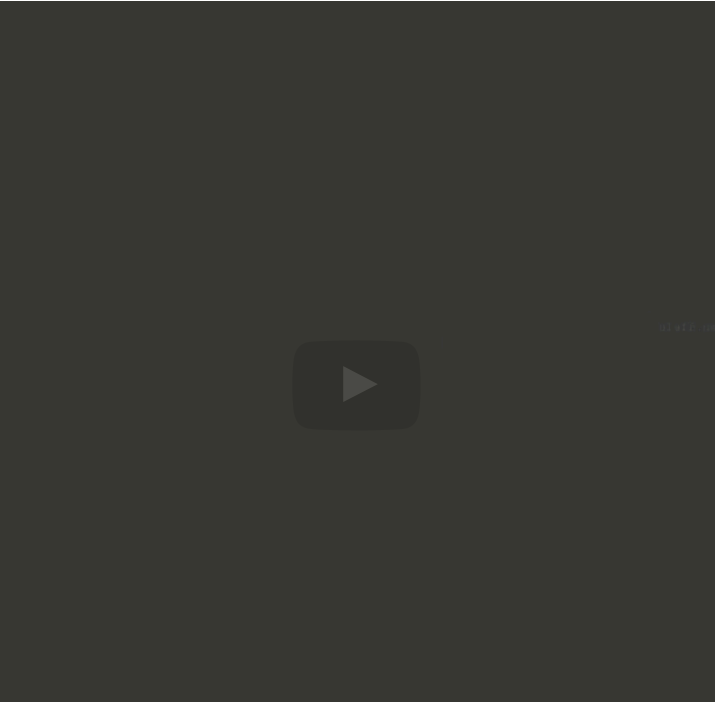
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it also says start with a matrix of V equal to I . But now in each iteration

take your matrix A , compute its QR factorization, and once you have its QR factorization, multiply R times Q , creating a new

matrix A . This is no longer the same matrix as we started with. And then take this matrix Q and multiply it from the right times our original matrix V , or the V

that we're now accumulating. And that becomes our

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Reading assignment

1/1 point (graded)



Read Unit 10.2.1 of the notes. [\[LINK\]](#)

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Homework 10.2.1.1

1/1 point (graded)
Show that for the algorithm on the right
 $A^{(k+1)} = Q^{(k+1)H} A^{(k)} Q^{(k+1)}.$

☒ done



Solution

For a complete solution, see the [notes](#).

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Homework 10.2.1.2

1/1 point (graded)
In the above algorithms, for all k ,

- $\hat{A}^{(k)} = A^{(k)}.$
- $\hat{R}^{(k)} = R^{(k)}.$
- $\hat{V}^{(k)} = V^{(k)}.$

☒ done



Solution

For a complete solution, see the [notes](#).

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Homework 10.2.1.3

1/1 point (graded)
In the above algorithms, show that for all k

- $V^{(k)} = Q^{(0)} Q^{(1)} \dots Q^{(k)}.$
- $A^k = V^{(k)} R^{(k)} \dots R^{(1)} R^{(0)}.$ (Note: A^k here denotes A raised to the k th power.)

☒ done



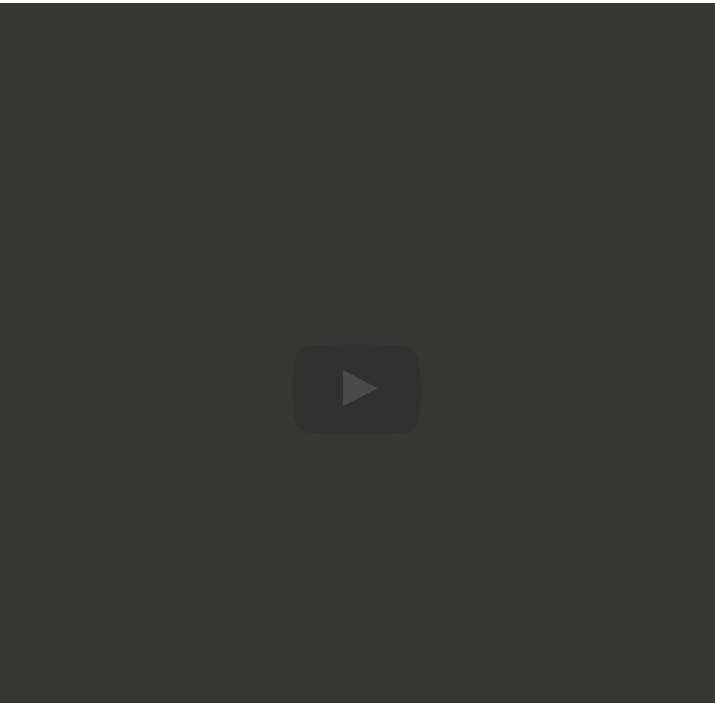
Solution

For a complete solution, see the [notes](#).

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becomes the diagonal matrix. And if you accumulate all of these transformations of the basis that you encounter along the way, what you end up with is a matrix where the columns of that matrix equal the eigenvectors associated with the eigenvalues that appear on the diagonal of matrix A. And that's how the two are related to each other. But this is surprisingly simple. Compute the QR factorization of A

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Homework 10.2.1.4

1/1 point (graded)
Examine [Assignments/Week10/matlab/SubspaceIterationAllVectors.m](#), which implements the subspace iteration in Figure 10.2.1.1 (left). Examine it by executing the script in [Assignments/Week10/matlab/test_simple_QR_algorithms.m](#).

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Homework 10.2.1.5

1/1 point (graded)
Copy [Assignments/Week10/matlab/SubspaceIterationAllVectors.m](#) into `SimpleQRAlg.m` and modify it to implement the algorithm in Figure 10.2.1.1 (right) as

```
function [ Ak, V ] = SimpleQRAlg( A, maxits, illustrate, delay )
```

Modify the appropriate line in [Assignments/Week10/matlab/test_simple_QR_algorithms.m](#), changing (0) to (1), and use it to examine the convergence of the method.

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